CECS 385
Population, Sampling, Bias, and Missingness

Overview

Use the provided **GSS_edited_values.xlsx** file to practice defining a population and sample, examining distributions, identifying missingness, and writing conclusions that match the limits of available data.

Important Note: This is an **edited subset of GSS** data containing 1,500 records and 24 variables. The original data is hosted https://gss.norc.org. The spreadsheet provided does not document the original GSS year, full sampling frame, survey weights, stratification, clustering, or data-collection design. As such, do not claim that your calculated percentages represent all U.S. adults. Your conclusions must be limited to this edited dataset.

Data and Data Dictionary

Open the GSS_edited worksheet in GSS_edited_values.xlsx. The Key worksheet contains the available variable names and labels. The data file has 1,500 rows and these 24 variables:

Variable	Description
age	Age of respondent
sex	Respondent's sex
richwork	If rich, continue or stop working
satjob	Job satisfaction
life	Whether life is exciting, dull, or routine
impjob	Importance of having a fulfilling job
hrs1	Hours worked last week
maeduc	Mother's highest year of school completed
paeduc	Father's highest year of school completed
satjob2	Job satisfaction (alternate item)
marital	Marital status
agewed	Age at first marriage
zodiac	Astrological sign
postlife	Belief in life after death
happy	General happiness
hapmar	Happiness of marriage
jobinc	Importance of a high income in a job
classicl	Attitude toward classical music
opera	Attitude toward opera
country	Attitude toward country-western music
rincom91	Respondent income category
tvhours	Hours of television watched
finrela	Self-rated financial situation
rincmdol	Income in dollars / numeric income field

Required Deliverables

Submit your report containing all required tables, charts, calculations, and written responses. Include your name and clearly label each part. You may use Excel, Google Sheets, R, Python, Tableau, etc. Explain how each table/chart is generated. Show enough work that another person could understand how you generated responses to each part.

Part 1: Population, Sample, and Unit of Analysis (15 points)
Write approximately 100–150 words responding to the following prompts.

1. What population does the original GSS generally intend to describe?
2. What is the unit of analysis in this worksheet?
3. What does one row represent?
4. What is the sample for your classroom analysis?
5. What is the likely sampling frame for the edited file? Explain what cannot be determined from the spreadsheet alone.
6. Explain why the 1,500 records should not automatically be treated as a simple random sample of all U.S. adults.
7. State one conclusion that would be inappropriate to make using this edited subset.

Part 2: Sample Distribution (25 points)
Create frequency tables for the variables below. For each table, report count and percentage.

1. age
2. sex
3. marital
4. happy
5. rincom91

Age Groups
Create an age_group variable using these categories:

Age group	Range
18–24	18–24
25–34	25–34
35–44	35–44
45–54	45–54
55–64	55–64
65–74	65–74
75+	75 or older

Use age_group, rather than every individual age, in your age frequency table and chart.

Visualizations
Create two visualizations:

1. A bar chart for a categorical variable, such as sex, marital, or happy.
2. A histogram of tvhours or a grouped bar chart for age_group.

Interpretation
Write approximately 150–200 words answering:

- Which groups are most and least represented in this edited sample?
- Which distributions are uneven, skewed, or affected by non-substantive responses?
- Does a sample of 1,500 records guarantee exact representation of every subgroup? Why not?
- What can and cannot be inferred from these distributions?

Part 3: Missingness and Non-Substantive Responses (20 points)
Examine missingness for these variables:

- rincom91
- satjob
- happy
- postlife
- tvhours

Step 1: Define Missingness

For this part, treat the following values as non-substantive / item-missing when they occur:

- Don't know or DK;
- No answer;
- Refused;
- blank or empty-string values;
- system-missing values;
- Not Available.

Treat Not applicable separately. It may represent a legitimate skip pattern rather than accidental missingness. For example, job satisfaction may be not applicable to people who were not working.

Step 2: Create a Missingness Table

Variable	Valid substantive responses	Not applicable	Don't know / refused / no answer / blank	System missing	Total records	Non-substantive missing %
Income (rincom91)					1,500	
Job satisfaction (satjob)					1,500	
General happiness (happy)					1,500	
Life after death (postlife)					1,500	
Television hours (tvhours)					1,500	

For tvhours, explain how you handled numeric missing values and whether any values appear implausible.

Step 3: Compare Missingness Across Groups

Choose one outcome variable, either rincom91 or satjob, and compare its missingness across:

- sex, or
- age_group, or
- marital.

Create a table showing each group's total count, missing-response count, and missing-response percentage.

Step 4: Interpretation

Write approximately 150–200 words addressing:

- Which variables have the most missing or non-substantive responses?
- Does missingness appear equally distributed across your chosen groups?
- Can these data establish whether missingness is random? Why or why not?
- Explain one way that deleting every record with a missing value could bias a result.

Part 4: Sampling Bias and Measurement Bias Assessment (15 points)
Write approximately 100–150 words evaluating possible limitations of the edited classroom dataset. Address the following:

- Coverage and Sampling-Frame Limits: Why is it a limitation that the original sampling frame and survey year are not documented in the edited spreadsheet?
- Nonresponse and Item Nonresponse: Use one missingness result from Part 3 as evidence and explain how people who decline to answer income or job questions might differ from those who answer.
- Measurement and Response Bias: Identify one variable that may be affected by social desirability, recall error, ambiguous wording, or respondent interpretation.Explain why the observed value may not perfectly measure the intended concept.
- Edited-File Limits: Explain why the classroom subset should not automatically be treated as a nationally representative sample. State one conclusion that is appropriate and one that is inappropriate.

Part 5: Descriptive Research Question (25 points)
Choose one question that can be explored descriptively with the data. Examples:

- How does general happiness (happy) vary across marital-status categories (marital)?
- How does self-rated financial situation (finrela) vary across income categories (rincom91)?
- How do attitudes toward classical music (classicl) vary by age group?
- How does the importance of a high income in a job (jobinc) vary by sex?
- How does television viewing (tvhours) vary across general-happiness categories?

Create one relevant table, chart, or cross-tabulation. Then write approximately 200–250 words that:

1. states the question;
2. reports the most relevant descriptive finding;
3. describes how missing or non-substantive responses were handled;
4. avoids causal language;
5. states at least two limitations; and
6. limits the conclusion to the edited classroom dataset.

Generative AI Use and Disclosure
If you use a generative-AI tool at any stage of this assignment, include a brief disclosure in your submission that identifies the tool and version, date used, purpose of use, and the prompts used or a concise description of them. For example, explain whether the tool was used for editing prose, explaining code, debugging an Excel formula, or interpreting a chart. You remain responsible for completing the analysis, verifying all calculations and code, checking that any AI-generated statement matches your actual results, correcting errors or fabricated information, and writing conclusions that accurately reflect this edited dataset. Do not submit AI-generated text, tables, charts, citations, or interpretations as your own work without disclosure. If you did not use generative AI, include the statement: **“I did not use generative AI for this assignment.”**